

Member ID: _____

Time: _____

Rank: _____



MANAGERIAL ACCOUNTING (135)

NATIONAL 2025

CONCEPT KNOWLEDGE:

Multiple Choice (10 @ 2 points each) _____ (20 points)

APPLICATION KNOWLEDGE:

Short Answer _____ (30 points)

Problem 1: Budgeting Problems _____ (50 points)

Problem 2: Capital Budgeting _____ (30 points)

TOTAL POINTS _____ ***(130 points)***

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

The test is divided into two parts, and you **MUST** complete both parts. You will have 90 minutes to complete your work.

PART ONE Instructions: Concept Knowledge

1. Open Google Chrome and navigate to compete.bpa.org.
2. Click on the Secondary Division Competitions button.
3. Enter your BPA Member ID (8-digits) and Password to access your test questions.
4. Answer all questions and submit your test.
5. Move on to Part Two.

PART TWO: Application Knowledge

Complete the questions in the rest of this test booklet by filling in the PDF.

EXAM GUIDELINES:

You have been hired as a Financial Assistant and will be keeping the accounting records for Digital Solutions, located at 700 Morse Road, Suite 201, Columbus, Ohio 43214. Digital Solutions provides accounting and other financial services for clients. You will complete jobs for Digital Solutions' own accounting records, as well as for clients.

You will have 90 minutes to complete your work. The test is divided into two parts: concept knowledge and application knowledge.

Your name and/or school name should *not* appear on any work you submit for grading. Write your Member ID in the provided space. Staple all pages in order before you turn in your test.

Assumptions to make when taking this assessment:

- Round all calculations to two decimals at the final step.
- Round all percentages to one decimal place.
- Use 360 days for interest calculations.

Short Answer (30 Points)

1. Bernie Brewer Sports Co. has several lines of baseball gloves for all levels of athletes ranging from tee-ball gloves to gloves used by professional athletes. The company has found it beneficial to split operations into two divisions based on the customer: Professional Athletes and Recreation. The following divisional information is available for the past year:

	Net Sales Revenue	Operating Income	Average Total Assets
Professional Athletes	\$ 7,500,000	\$ 450,000	\$ 2,500,000
Recreation	5,243,600	975,800	6,500,000

Bernie's management has specified a 13% target rate of return.

	ROI	Residual Income
Professional Athletes Division		
Recreation Division		

2. Royals Systems is the leading manufacturer of pitching machines. They have budgeted three hours of direct labor per pitching machine, at a standard cost of \$17 per hour. During August, employees worked 189 hours, completing 84 pitching machines. Royals paid the employees \$18 per hour.

Calculate the following variances and identify if each was favorable (F) or unfavorable (U).

Labor Rate Variance	
Labor Efficiency Variance	
Total Labor Spending Variance	

3. Dodger Industries manufactures batting helmets. During the second quarter of the year, the company manufactured 3,500 helmets, using 2,250 kilograms of plastic. The plastic cost the company \$17,100. According to the standard cost, each helmet should require 0.6 kg of plastic at a cost of \$8 per kilogram.

Calculate the following variances and identify if each was favorable (F) or unfavorable (U).

Materials Price Variance	
Materials Quantity Variance	
Total Materials Spending Variance	

Problem 1: Budgeting Problems (50 Points)

- A. Astros Company makes a product that is very popular as a Christmas gift. Thus, peak sales occur in December of each year, as shown in the sales budget for the fourth quarter given below:

	October	November	December	Total
Budgeted Sales (all on account)	\$200,000	\$300,000	\$500,000	\$1,000,000

From past experience, the company has learned that 20% of a month's sales are collected in the month of sale, 70% are collected in the month following sale, and the remaining 10% are collected in the second month following sale. Bad debts are negligible and can be ignored.

How much cash is expected to be collected in December?	
What will be the balance in Accounts Receivable at the end of December?	

- B. Red Sox Company has budgeted sales of its popular batting gloves for the next four months as follows:

	Unit Sales
January	50,000
February	75,000
March	90,000
April	80,000

The company is preparing a production budget for the first quarter. Past experience has shown that end-of-month inventory levels must equal 10% of the following month's unit sales. The inventory at the end of December was 5,000 units.

How many units should the company plan to produce in January?	
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- C. Each baseball that Tiger Corp. produces requires 3 grams of leather. The leather costs \$11.50 per gram. Budgeted production of baseballs is given below for the last quarter of Year 2 and the first quarter of Year 3.

	Year 2			Year 3
	October	November	December	January
Budgeted Production	6,000	9,000	15,000	7,000

The inventory of leather at the end of the quarter must be equal to 20% of the following month's quarter production needs. Tiger estimates that 3,600 grams of leather will be on hand at the start of October.

How many grams of leather should be purchased in October?	
How much should Tiger budget for October materials purchases?	
The company pays for 60% of the purchases in the month of purchase, and 40% in the month following purchase. Accounts Payable at the end of September is \$66,000. How much will the cash payments be for October?	
What will be the balance in Accounts Payable at the end of October?	

- D. The MN Twins expect to produce 8,000 team jerseys in the next quarter. Each unit requires 1.35 direct labor hours, and each employee is paid \$12.00 per hour.

How many hours of direct labor should Twins budget for?	
How much direct labor cost should Twins budget for?	

- E. Mr. Met Inc. has a cash balance of \$20,000 on April 1. The company is now preparing the cash budget for the second quarter. Budgeted cash collections and payments are as follows:

	Apr	May	Jun
Cash collections	\$220,000	\$225,000	\$240,000
Cash payments:			
Purchases of direct materials	44,000	45,000	46,800
Payments for Direct Labor	56,000	54,600	57,000
MOH Expenses (including \$15,000 of Depreciation)	83,000	87,500	89,000
Selling & Administrative Expenses (including \$5,000 of depreciation)	66,000	63,000	69,000

If Mr. Met. wants to have a minimum cash balance of \$20,000 at the end of each month. How much will he need to borrow at the end of April?	
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Problem 2: Capital Budgeting (30 Points)

Mariner Moose Inc. is considering a capital investment of \$261,456. They are considering purchasing new machinery that is expected to have a useful life of 5 years with no salvage value. Depreciation is calculated using the straight-line method. During the life of the investment, annual net income is expected to be \$25,709 and annual net cash inflows are expected to be \$78,000. Mariner Moose has a 10% minimum acceptable rate of return on all investments.

What is the payback period of the investment?	
What is the net present value of the investment?	
What is the approximate internal rate of return of the investment?	

Present value interest factor of \$1 per period at i% for n periods, PVIF(i,n).																				
Period	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065
16	0.853	0.728	0.623	0.534	0.458	0.394	0.339	0.292	0.252	0.218	0.188	0.163	0.141	0.123	0.107	0.093	0.081	0.071	0.062	0.054
17	0.844	0.714	0.605	0.513	0.436	0.371	0.317	0.270	0.231	0.198	0.170	0.146	0.125	0.108	0.093	0.080	0.069	0.060	0.052	0.045
18	0.836	0.700	0.587	0.494	0.416	0.350	0.296	0.250	0.212	0.180	0.153	0.130	0.111	0.095	0.081	0.069	0.059	0.051	0.044	0.038
19	0.828	0.686	0.570	0.475	0.396	0.331	0.277	0.232	0.194	0.164	0.138	0.116	0.098	0.083	0.070	0.060	0.051	0.043	0.037	0.031
20	0.820	0.673	0.554	0.456	0.377	0.312	0.258	0.215	0.178	0.149	0.124	0.104	0.087	0.073	0.061	0.051	0.043	0.037	0.031	0.026
25	0.780	0.610	0.478	0.375	0.295	0.233	0.184	0.146	0.116	0.092	0.074	0.059	0.047	0.038	0.030	0.024	0.020	0.016	0.013	0.010
30	0.742	0.552	0.412	0.308	0.231	0.174	0.131	0.099	0.075	0.057	0.044	0.033	0.026	0.020	0.015	0.012	0.009	0.007	0.005	0.004
35	0.706	0.500	0.355	0.253	0.181	0.130	0.094	0.068	0.049	0.036	0.026	0.019	0.014	0.010	0.008	0.006	0.004	0.003	0.002	0.002
40	0.672	0.453	0.307	0.208	0.142	0.097	0.067	0.046	0.032	0.022	0.015	0.011	0.008	0.005	0.004	0.003	0.002	0.001	0.001	0.001
50	0.608	0.372	0.228	0.141	0.087	0.054	0.034	0.021	0.013	0.009	0.005	0.003	0.002	0.001	0.001	0.001	0.000	0.000	0.000	0.000

Present value interest factor of an (ordinary) annuity of \$1 per period at i% for n periods, PVIFA(i,n).																				
Period	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675
16	14.718	13.578	12.561	11.652	10.838	10.106	9.447	8.851	8.313	7.824	7.379	6.974	6.604	6.265	5.954	5.668	5.405	5.162	4.938	4.730
17	15.562	14.292	13.166	12.166	11.274	10.477	9.763	9.122	8.544	8.022	7.549	7.120	6.729	6.373	6.047	5.749	5.475	5.222	4.990	4.775
18	16.398	14.992	13.754	12.659	11.690	10.828	10.059	9.372	8.756	8.201	7.702	7.250	6.840	6.467	6.128	5.818	5.534	5.273	5.033	4.812
19	17.226	15.678	14.324	13.134	12.085	11.158	10.336	9.604	8.950	8.365	7.839	7.366	6.938	6.550	6.198	5.877	5.584	5.316	5.070	4.843
20	18.046	16.351	14.877	13.590	12.462	11.470	10.594	9.818	9.129	8.514	7.963	7.469	7.025	6.623	6.259	5.929	5.628	5.353	5.101	4.870
25	22.023	19.523	17.413	15.622	14.094	12.783	11.654	10.675	9.823	9.077	8.422	7.843	7.330	6.873	6.464	6.097	5.766	5.467	5.195	4.948
30	25.808	22.396	19.600	17.292	15.372	13.765	12.409	11.258	10.274	9.427	8.694	8.055	7.496	7.003	6.566	6.177	5.829	5.517	5.235	4.979
35	29.409	24.999	21.487	18.665	16.374	14.498	12.948	11.655	10.567	9.644	8.855	8.176	7.586	7.070	6.617	6.215	5.858	5.539	5.251	4.992
40	32.835	27.355	23.115	19.793	17.159	15.046	13.332	11.925	10.757	9.779	8.951	8.244	7.634	7.105	6.642	6.233	5.871	5.548	5.258	4.997
50	39.196	31.424	25.730	21.482	18.256	15.762	13.801	12.233	10.962	9.915	9.042	8.304	7.675	7.133	6.661	6.246	5.880	5.554	5.262	4.999